

THE AI TOOLBOX · CLASS 07

What Is Data? How to Obtain It

Understand where data comes from — and **judge it before you trust it.**

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

The research stack

Elicit · Semantic Scholar · Connected Papers
— tools find papers, **you** read the five that matter.

Citation graphs

Back to the roots, forward to the frontier; gaps between clusters are opportunities (Swanson!).

Your literature map

5 papers + who-builds-on-whom — your question now has a neighborhood.

Homework share: show your literature map and your GitHub repo README. Which paper is closest to your question? **Today that question goes hunting for its data** — Stage 3 of the pipeline starts now.

Where we're going

- **0–8'** Homework share + the car that built a benchmark
- **8–18'** What data *is* · structured vs. unstructured
- **18–26'** Quantitative vs. qualitative — why the type picks the tools
- **26–38'** Where data comes from · crawlers & APIs · licenses & bias
- **38–46'** Demos: Kaggle datasets · Roboflow Universe · ScholarMind
- **46–54'** Two labs: pixels → features · judge four datasets
- **54–60'** Your dataset datasheet, homework & wrap-up
- Quizzes &  comments throughout

By the end: tell structured from unstructured data, name the places researchers get data, check a license before trusting a dataset, and document yours like a professional.

WARM-UP · ACTIVATE

Quick check · tap an answer

You want to study night-time pedestrian detection. Thousands of labelled night-driving images already exist. What are the TWO questions to ask before using them?

How big is the file, and will it fit on my laptop?

Who collected this data and how — and does its license allow me to use it?

Is it popular, and does it have a cool name?

None — public data on the internet is free to use however I want

THE STORY

They drove a city to build a benchmark

In 2012, researchers in **Karlsruhe, Germany** bolted cameras, lasers, and GPS onto a station wagon and drove it around the city for hours. The result — the **KITTI dataset** — became the testing ground that self-driving perception research used **for the next decade**. Thousands of papers were built on data from one rigged-up car driving real streets.



Speak this story

Neural voice · click to play / pause

Sound familiar? It's the ImageNet lesson (Class 3) on wheels: the field didn't advance because someone had a cleverer model — it advanced because someone **collected and shared the right data**. Datasets are research infrastructure.

CONCEPT 1 · LOOK AROUND YOU

Everything is data (once you record it)



Your height & weight?

Data. Quantified body measurements — numbers with units.



A social-media post?

Data. Text carries opinions; likes and reposts record relationships.



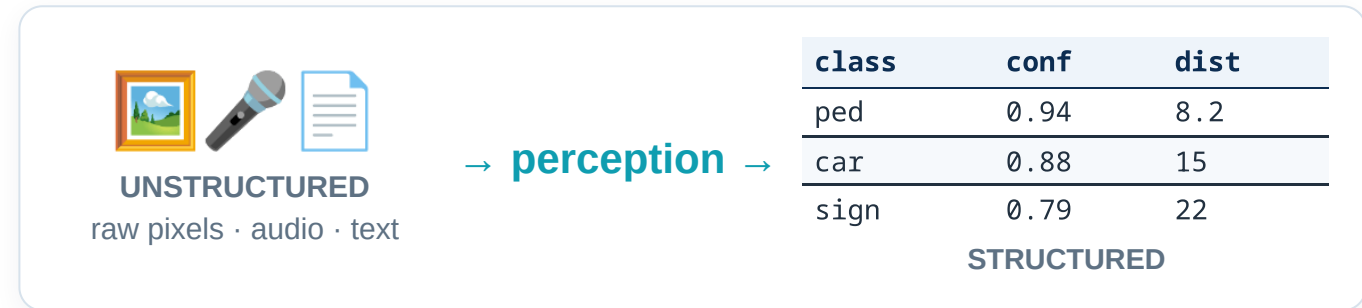
A phone photo?

Data. Millions of pixel values — a grid of numbers representing light.

KEY IDEA — data is a **recorded symbol of something real**: numbers, text, pixels, sounds. And AI's whole trick (Class 1!) is learning patterns from those symbols. No recording → no data → nothing to learn from. That's why researchers obsess over *collection*.

Structured vs. unstructured

- **Structured** — fits a table: rows × columns, ready for pandas (Class 5's whole world).
- **Unstructured** — images, audio, video, free text: no rows and columns yet.
- **Perception is the bridge:** a model turns raw pixels into a structured table of detections.



Most of the world's exciting AI data starts unstructured — that's exactly why deep learning matters (Class 1's DL ring). Lab 1 today: you'll perform this pixels → table transformation yourself, by hand.

Quantitative vs. qualitative — the type picks the tools



Quantitative — "how much"

Discrete: counts — objects per frame, number of siblings.

Continuous: any value in a range — distance_m, temperature, confidence.



Qualitative — "what kind"

Nominal: categories, no order — pedestrian / car / sign, blood type.

Ordinal: ordered, uneven gaps — grades A/B/C, satisfaction ★–
★★★★★.

Why care? The type decides everything downstream: which **chart** (Class 9), which **statistic** (mean of nominal = nonsense), and which **encoding** a model needs (Class 8 turns "pedestrian" into numbers — carefully). Misread the type, and every later stage quietly breaks.

CHECK · CLASSIFY IT

Quick check · tap an answer

Your study-tracker table has a column 'focus level' recorded as ★ to ★★★★★. What kind of data is that?

Continuous quantitative — it's a number from 1 to 5

Nominal qualitative — just five categories

Ordinal qualitative — ordered levels, but the gaps between stars aren't equal units

Unstructured — stars aren't data

Where researchers actually get data



Public datasets

Kaggle, Roboflow Universe, UCI, Hugging Face, government portals, academic benchmarks (KITTI!). Free, labeled, documented — **start here**, always.



Your own / proprietary

Data you collect (surveys, sensors, experiments) or a company's internal records. Highest relevance, highest cost — collection is a project in itself.



The open web

Crawlers scrape pages; **APIs** hand you structured data politely. Fresh and vast — but with rules (next slide).

For this course: public datasets are your well — your Class-5 skills work on them immediately, and their licenses are written down. Well-chosen public data beats painfully self-collected data for a first project, every time.

Crawlers scrape; APIs ask nicely



Web crawler

A script that fetches pages and parses out the data (Python: `requests` + `BeautifulSoup`). Powerful — and easy to do rudely or illegally.



API

The site's official data door: get a key → read the docs → request → receive clean JSON. Stable, legal, structured — **always prefer the API if one exists** (weather, maps, social platforms...).



The rulebook (non-negotiable): respect `robots.txt` · throttle your request rate · never collect personal or private information · check the site's terms and data copyright. A crawler that ignores these isn't research — it's trespassing with extra steps.

Every dataset has a license and a story



The license

CC0 / CC-BY: use freely (credit for BY).

Research-only: no commercial use.

Unknown: treat as *unusable*. The license is usually one click away — read it.



The collection story

Who collected it, when, where, how? KITTI = one car, one German city, daytime, 2012. That story defines what the data can — and cannot — say.



The baked-in bias

KITTI's story means: German roads, good weather, 2012-era cars. A model trained on it has **never seen** a snowy night in Shanghai. Bias isn't a scandal — it's a *documented limitation*... if you look.

The instructor's rule: if the license is unclear — **pick a different dataset. Don't guess.** And write the collection story into your datasheet: it's tomorrow's "limitations" section (Class 4!) writing itself.

Quick check · tap an answer

You find the perfect dataset for your project, but its page shows no license at all. What's the researcher's move?

Use it — no license means no rules

Use it but don't tell anyone where the data came from

Treat it as unusable: pick a licensed alternative, or write the author to ask

Copy it to your own page and add a license yourself

The datasets self-driving research runs on

KITTI / nuScenes

The classic benchmarks: camera + lidar + labeled boxes. KITTI (2012) started it; nuScenes added nights, rain, and 360° sensors — *because KITTI's story lacked them*.

Waymo Open

Industrial scale: ~2,000 hours, multi-sensor, many cities. Free for research — **license explicitly non-commercial**. Modern papers benchmark here.

Roboflow Universe

Thousands of small, ready-to-train sets (including night-pedestrian ones!) — the practical shelf for a student project: download in YOLO format, train same-day.

Read the shelf like a researcher: each newer dataset exists to fix the previous one's **collection-story gaps** — nuScenes added the nights KITTI lacked. Dataset evolution IS the field saying "our data was biased; here's better." Prefer another domain? Same three-tier shelf exists everywhere.

DISCUSS · 5 MINUTES · YOUR DREAM DATASET

Who has your data — and would they share it?

Name the dataset your **dream project** would need. Not a course project — the real dream.

Trace it

Who would have collected it? Why did they collect it? Would they let *you* use it — and under what license?

Stress-test it

Structured or unstructured? And what bias is baked into *how* it was collected — who's missing from it?

Pairs, 3 minutes — then a few dreams out loud. Post yours in .

LIVE DEMO 1

Kaggle Datasets — read before you download

Not the competitions this time — the dataset library. Watch the researcher's reading order: license first, dictionary second, download last.

You ▶ `kaggle.com/datasets` → search your topic (try: pedestrian detection / air quality / your domain)

You ▶ On a result: find the LICENSE field first – CC0? CC-BY? something scarier?

You ▶ Read the data dictionary: every column's meaning and units (Class 5's dream)

You ▶ Check 'Usability' score + the discussion tab: what problems have others hit?



kaggle.com/datasets

LIVE DEMO 2

Roboflow Universe — images and labels, side by side

For unstructured (vision) data: browse actual images WITH their boxes drawn — see exactly what 'labeled' means before committing.

You ▶ `universe.roboflow.com` → search: night pedestrian (yes, it exists – several sets)

You ▶ Open one → browse images: are the boxes accurate? tight? consistent?

You ▶ Check: image count · classes · license · when it was updated

You ▶ Note the download formats (YOLO, COCO...) – this plugs straight into later classes



universe.roboflow.com

LIVE DEMO 3

ScholarMind — does data exist for YOUR question?

Our platform's dataset search covers Hugging Face + UCI + Kaggle at once — the fastest reality-check for a research question.

You ▶ Enter your research topic → it proposes questions grounded in real datasets

You ▶ For your Class-2 strong question: which datasets does it surface? Any surprises?

You ▶ Compare two candidates: size, labels, license – which passes today's judgment layer?

You ▶ No dataset found? That's pitfall #4 (Class 3) knocking – adjust the question NOW, not in week 6.



academic.hopeembark.org

Turn unstructured pixels into a structured row

An image is just numbers. Here's an 8×8 "photo" — **you** play the perception model: extract features, produce the structured row. Concept 2, by hand.

```
1 # An "image" is just numbers: 8x8 grayscale, 0 = dark ... 9 = bright
2 image = [
3     [0,0,0,9,9,0,0,0],
4     [0,0,9,9,9,9,0,0],
5     [0,0,9,9,9,9,0,0],
6     [0,0,0,9,9,0,0,0],
7     [0,0,0,9,9,0,0,0],
8     [0,0,9,9,9,9,0,0],
9     [0,9,9,0,0,9,9,0],
10    [0,9,0,0,0,0,9,0],
11 ] # squint: a tiny pedestrian silhouette
12
```



Run



Explain



Debug



Improve



Assistant

Python loads on first Run

► Run the code to see the output here.

Four candidate datasets — which would a researcher use?

Today's checklist as a script: license, labels, size, collection story. Four candidates for the night-pedestrian project — score them, then **argue with the scores**.

```
1 # The datasheet checklist, as code. 4 candidates for a night-ped study:
2 datasets = [
3     {"name": "NightPeds-500", "license": "CC-BY 4.0", "labeled": True,
4      "size": 500, "collection": "dashcam, 3 cities, winter nights"},
5     {"name": "MegaDrive-1M", "license": "unknown", "labeled": True,
6      "size": 1000000, "collection": "scraped from video sites"},
7     {"name": "SunnyStreets", "license": "CC0", "labeled": False,
8      "size": 8000, "collection": "one campus, daytime only"},
9     {"name": "CampusWalk", "license": "research-only", "labeled": True,
10      "size": 120, "collection": "10 students, one evening"},
11 ]
12
```



Run



Explain



Debug



Improve



Assistant

Python loads on first Run

► Run the code to see the output here.

YOUR PROJECT

Choose and document your dataset

Pick the dataset your project will actually use — and document it the way a researcher would, in a **datasheet**.

- Browse Kaggle / Roboflow Universe for candidates matching your question
- Check the **license** and the size — apply the judgment layer
- Record the **source and collection story** (who, when, how)
- Describe **every variable**: meaning, type (today's vocabulary!), units

Deliverable

A dataset datasheet: source, license, size, collection story, and a variable table.

Success looks like

License and source clearly stated — and every variable has a meaning and units.

BUILD IT NOW · HANDS-ON

Hunt, judge, document

1. Search Kaggle Datasets and Roboflow Universe for your question's data (ScholarMind if stuck)
2. Shortlist 2 candidates → run the **judgment layer**: license · labels · size · collection story
3. Pick one. Download it (or note the access link)
4. Fill the datasheet: source · license · size · collection story · **variable table** (name / meaning / type / units)
5. Load it with Class-5 verbs — does `df.head()` match the dictionary's promises?

Kaggle

Roboflow Universe

ScholarMind

Colab

Instructor's rule, again: license unclear → different dataset. And step 5 catches the classic surprise: documentation says one thing, the actual columns say another. Trust `df.head()`.

What a finished datasheet looks like

Dataset: "Night-Time Pedestrians v2" (Roboflow Universe)

Source: universe.roboflow.com/... · **License:** CC-BY 4.0 ✓ · **Size:** 1,847 images, 4,210 boxes

Collection story: dashcam footage, 2 US cities, Nov–Feb, dusk-to-night only; boxes drawn by the uploading team.

Variables: image (unstructured, 640×640 px) · class (nominal: pedestrian/cyclist) · x,y,w,h (continuous, 0–1 normalized box) · lighting (ordinal: dusk/night/deep-night)

Known limitations (from the story): US cities only · winter clothing bias · no rain or snow nights · single labeling team, no cross-check. *Claims about other conditions are extrapolation.*

The bar: every field filled from evidence, variable types named with today's vocabulary, and the limitation box **derived from the collection story** — not guessed. This page is the first section of your final paper's "Data" chapter, already written.

Today's vocabulary

Structured / unstructured — fits a table now vs. needs a model (perception) to get there.

Qualitative (nominal / ordinal) — "what kind": unordered categories vs. ordered levels.

API — a site's official structured-data door; prefer it over scraping.

Quantitative (discrete / continuous) — "how much": counts vs. measurements.

License (CC0 / CC-BY / research-only) — the legal terms of use; unknown = unusable.

Datasheet — a dataset's documentation: source, license, collection story, variables, limitations.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your datasheet** and submit it in Google Classroom — and commit it to your GitHub repo from Class 6 (it's your project's first real artifact).
2. **Name one bias** your chosen dataset inherits from its collection story — one honest sentence in the datasheet's limitation box.
3. **Load it:** open your dataset in Colab, run the four verbs, and screenshot `df.head()` + `df.info()`. Do the columns match your variable table?

★ Stretch (optional)

Skim the original **KITTI paper's** dataset section ("Are we ready for autonomous driving?") — watch professionals write a datasheet at full scale: sensors, calibration, and admitted limitations.

[cvlibs.net · KITTI](https://www.cvlibs.net/datasets/kitti/)

Task 3 matters most: next class starts from **your loaded dataset** — Class 8 cleans exactly what you chose today. Questions → 

TODAY'S LINE TO REMEMBER

"Data is the new oil."

— Clive Humby (2006)

...and Humby's next line is the part everyone forgets: like oil, it's **valueless until refined**. Refining is next week.



Speak this

WRAP-UP

Three things to keep

- Structured fits a table; unstructured doesn't (yet) — **perception is the bridge**
- Data has a **license and a collection story** — judge both before you trust it
- A **datasheet** documents your data: source, variables, and honest limitations

Exit ticket: name one possible bias in the dataset you chose today. Post it in .

Next · Class 8 — refining the oil: turn your messy dataset into something a model can learn from.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

The datasets that made AI

a short hall of fame

Scraping ethics, expanded

the rulebook with reasons

Bonus quiz

one more check

Datasets that changed AI

MNIST (1998) — 70k handwritten digits. The "hello world" every ML student still meets first.

ImageNet (2009) — 14M labeled images; ignited deep learning (Class 3's story).

KITTI (2012) — one station wagon, a decade of self-driving research (today's story).

COCO (2014) — objects in messy, real scenes; YOLO's proving ground (your Class-4 paper!).

The pattern: each one is remembered not for an algorithm but for **collection + labels + open sharing**. Fields advance when someone does the unglamorous data work — a career hint hiding in plain sight.

Why the crawler rules exist

robots.txt & rate limits

The site's stated rules + not overwhelming their servers. Ignoring them can take a small site down — that's harm, not research.

Copyright & terms

Text, images, and databases have owners; terms of service are agreements. "For research" is not a universal exemption.

Privacy

Public ≠ fair game: names, faces, and posts belong to people. Scraped personal data can't be "un-collected" — the harm outlives the project.

The honest test

Would you be comfortable telling the site's owner — and the people in the data — exactly what you collected and why? If not, stop.

Quick check · tap an answer

A 1,000,000-image dataset with an unknown license, versus a 500-image dataset with CC-BY and a documented collection story that matches your question. Which does the researcher pick?

The big one — a million images always wins

The small one: usable license + right collection story beat raw size

Neither; wait for a perfect dataset

Both; merge them and hope